

2 Lab Calculations

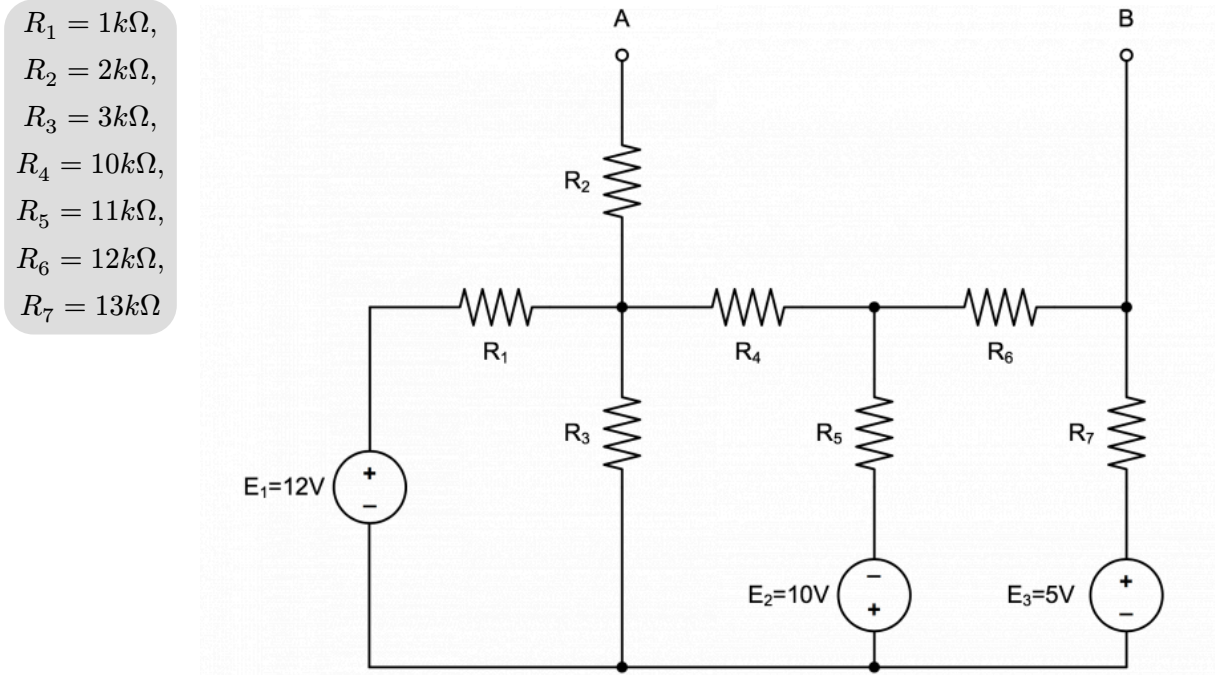
Section 13

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2.A Nodal & Mesh Analysis, Thevenin's Theorem and Superposition

For the circuit in Figure 3-1, all 7 resistors have different resistance values. For each resistor, choose any resistance value within the range of $1k\Omega$ to $56k\Omega$ unless specified otherwise.



2.A.1

Short AB, as shown in Figure 3-2(a). Use mesh analysis to calculate the voltage across each resistor and the current through AB, I_{AB} .

KVL@ i_1

$$12V = R_1 i_1 + R_3 (i_1 - i_2)$$

KVL@ i_2

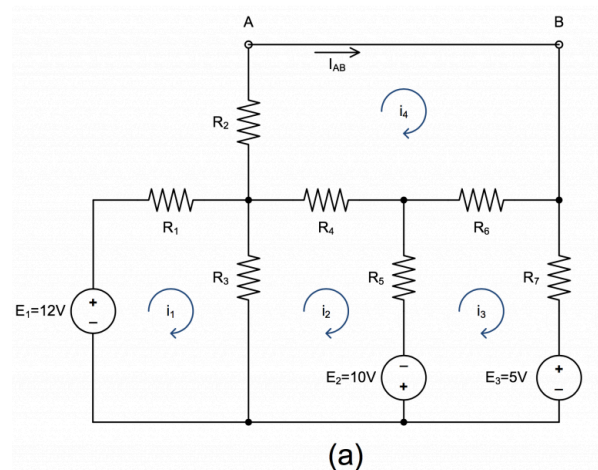
$$10V = R_3 (i_2 - i_1) + R_4 (i_2 - i_4) + R_5 (i_2 - i_3)$$

KVL@ i_3

$$-5V - 10V = R_5 (i_3 - i_2) + R_6 (i_3 - i_4) + R_7 i_3$$

KVL@ i_4

$$0 = R_4 (i_4 - i_2) + R_2 i_4 + R_6 (i_4 - i_3)$$



MATLAB code:

```
% Given values
R1 = 1e3; % Ohms
R2 = 2e3; % Ohms
R3 = 3e3; % Ohms
R4 = 10e3; % Ohms
R5 = 11e3; % Ohms
R6 = 12e3; % Ohms
R7 = 13e3; % Ohms

syms i1 i2 i3 i4;
% KVL equations
eq1 = 12 == R1*i1 + R3*(i1 - i2);
eq2 = 10 == R3*(i2 - i1) + R4*(i2 - i4) + R5*(i2 - i3);
eq3 = -5 - 10 == R5*(i3 - i2) + R6*(i3 - i4) + R7*i3;
eq4 = 0 == R4*(i4 - i2) + R2*i4 + R6*(i4 - i3);
% Solve for currents
currents = solve([eq1, eq2, eq3, eq4], [i1, i2, i3, i4])
```

$$\begin{aligned}i_1 &= 3.912mA \\i_2 &= 1.215mA \\i_3 &= 148\mu A \\i_4 &= 581\mu A \\i_{AB} &= i_4 = 581\mu A\end{aligned}$$

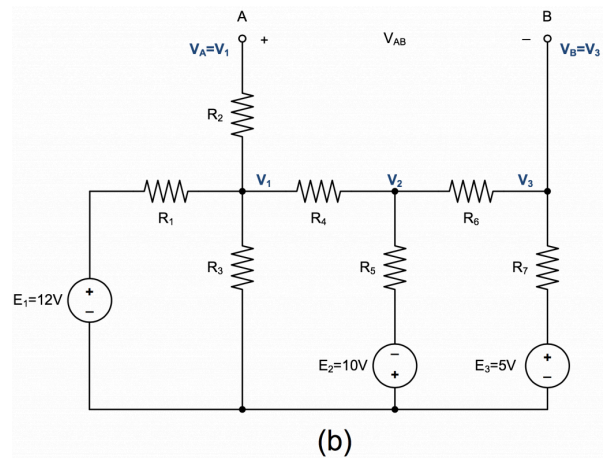
2.A.2

Leave AB open, as shown in Figure 3-2(b). Use nodal analysis to calculate the voltage across each resistor as well as the voltage across AB, V_{AB} .

$$\begin{aligned}&\text{KCL@}V_1 \\0 &= \frac{12V - V_1}{R_1} + \frac{0 - V_1}{R_3} + \frac{V_2 - V_1}{R_4}\end{aligned}$$

$$\begin{aligned}&\text{KCL@}V_2 \\0 &= \frac{V_1 - V_2}{R_4} + \frac{-10 - V_2}{R_5} + \frac{V_3 - V_2}{R_6}\end{aligned}$$

$$\begin{aligned}&\text{KCL@}V_3 \\0 &= \frac{V_2 - V_3}{R_6} + \frac{5 - V_3}{R_7}\end{aligned}$$



MATLAB Code:

```
% Given values
R1 = 1e3; % Ohms
R2 = 2e3; % Ohms
R3 = 3e3; % Ohms
R4 = 10e3; % Ohms
R5 = 11e3; % Ohms
R6 = 12e3; % Ohms
R7 = 13e3; % Ohms

syms V1 V2 V3;
% KCL equations
eq1 = 0 == (12 - V1)/R1 + (0 - V1)/R3 + (V2 - V1)/R4;
eq2 = 0 == (V1 - V2)/R4 + (-10 - V2)/R5 + (V3 - V2)/R6;
eq3 = 0 == (V2 - V3)/R6 + (5 - V3)/R7;
% Solve for node voltages
voltages = solve([eq1, eq2, eq3], [V1, V2, V3])
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$$V_1 = 8.287V$$

$$V_2 = 0.572V$$

$$V_3 = 2.698V$$

$$V_{AB} = V_1 - V_3 = 5.715V$$

2.A.3

Find Thevenin's and Norton's Equivalent using the results from the previous two steps.

$$V_{TH} = V_{AB} = 5.715V$$

$$R_{TH} = \frac{V_{AB}}{I_{AB}} = 9.836k\Omega$$

$$I_N = I_{AB} = 581\mu A$$

$$R_N = R_{TH} = 9.836k\Omega$$

2.A.4

Connect a resistor between nodes A and B. Use Thevenin's theorem to calculate the current through this resistor for the following 3 resistor values: $1k\Omega$, $2.2k\Omega$ and $4.7k\Omega$.

$$I_1 = \frac{V_{TH}}{R_{TH} + 1k\Omega} = 527\mu A$$

$$I_2 = \frac{V_{TH}}{R_{TH} + 2.2k\Omega} = 475\mu A$$

$$I_3 = \frac{V_{TH}}{R_{TH} + 4.7k\Omega} = 393\mu A$$

2.A.5

Leave AB open. Find the voltage across AB, V_{AB} for the following 3 cases:

Case 1: E_1 is turned on while both E_2 and E_3 are turned off.

$$\text{eq1} = 0 == (12 - V_1)/R_1 + (0 - V_1)/R_3 + (V_2 - V_1)/R_4;$$

$$\text{eq2} = 0 == (V_1 - V_2)/R_4 + (0 - V_2)/R_5 + (V_3 - V_2)/R_6;$$

$$\text{eq3} = 0 == (V_2 - V_3)/R_6 + (0 - V_3)/R_7;$$

$$V_1 = 8.633V$$

$$V_2 = 3.739V$$

$$V_3 = 1.944V$$

$$V_{AB} = V_1 - V_3 = 6.689V$$

Case 2: E_2 is turned on while both E_1 and E_3 are turned off.

$$\text{eq1} = 0 == (0 - V_1)/R_1 + (0 - V_1)/R_3 + (V_2 - V_1)/R_4;$$

$$\text{eq2} = 0 == (V_1 - V_2)/R_4 + (-10 - V_2)/R_5 + (V_3 - V_2)/R_6;$$

$$\text{eq3} = 0 == (V_2 - V_3)/R_6 + (0 - V_3)/R_7;$$

$$V_1 = -283mV$$

$$V_2 = -4.060V$$

$$V_3 = -2.111V$$

$$V_{AB} = V_1 - V_3 = 1.828V$$

Case 3: E_3 is turned on while both E_1 and E_2 are turned off.

$$\text{eq1} = 0 == (0 - V_1)/R_1 + (0 - V_1)/R_3 + (V_2 - V_1)/R_4;$$

$$\text{eq2} = 0 == (V_1 - V_2)/R_4 + (0 - V_2)/R_5 + (V_3 - V_2)/R_6;$$

$$\text{eq3} = 0 == (V_2 - V_3)/R_6 + (5 - V_3)/R_7;$$

$$V_1 = 62mV$$

$$V_2 = 893mV$$

$$V_3 = 2.864V$$

$$V_{AB} = V_1 - V_3 = -2.802V$$

Superposition:

Adding the three voltages demonstrates voltage superposition. The actual voltage can be calculated by summing the individual contributions from each voltage source. The $1mV$ discrepancy is due to accumulated rounding errors in the calculations.

$$\Sigma V_{AB} = 6.689V + 1.828V + (-2.802V) = 5.714 \approx 5.715V$$